27: Customer Balance Ranking

Aim

To write a COBOL program to read customer IDs and balances, sort the records in descending order of balance, arrange equal balances alphabetically by customer ID, and display the top three customers.

Algorithm

1. Start the program.
2. Read the number of customers `N`.
3. Read each customer's ID and balance.
4. Sort the records using bubble sort.
5. If balances are equal, compare customer IDs alphabetically.
6. Swap the complete customer records whenever required.
7. Display the first three records, or all records if `N` is less than three.
8. Stop the program.

Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. BALANCE-RANKING.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 N PIC 9(4).
01 I PIC 9(4).
01 J PIC 9(4).
01 T-ID PIC X(10).
01 T-BAL PIC 9(9).
```

```
01 CUST-TABLE.
  05 CUST OCCURS 1000 TIMES.
    10 C-ID PIC X(10).
    10 C-BAL PIC 9(9).
PROCEDURE DIVISION.
  ACCEPT N
  PERFORM VARYING I FROM 1 BY 1 UNTIL I > N
    ACCEPT C-ID(I)
    ACCEPT C-BAL(I)
  END-PERFORM
  PERFORM VARYING I FROM 1 BY 1 UNTIL I >= N
   PERFORM VARYING J FROM 1 BY 1 UNTIL J > N - I
    IF C-BAL(J) < C-BAL(J + 1) OR
      (C-BAL(J) = C-BAL(J + 1) AND
       C-ID(J) > C-ID(J + 1))
        MOVE C-ID(J) TO T-ID
        MOVE C-BAL(J) TO T-BAL
        MOVE C-ID(J + 1) TO C-ID(J)
        MOVE C-BAL(J + 1) TO C-BAL(J)
        MOVE T-ID TO C-ID(J + 1)
        MOVE T-BAL TO C-BAL(J + 1)
    END-IF
   END-PERFORM
  END-PERFORM
  PERFORM VARYING I FROM 1 BY 1
    UNTIL I > N OR I > 3
    DISPLAY FUNCTION TRIM(C-ID(I)) " " C-BAL(I)
  END-PERFORM
```

STOP RUN.

Sample input

4

C03

1000

C01

2000

C04

2000

C02

500

Output

```
C01 000002000
C04 000002000
C03 000001000
```